Jeet Shah

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Portfolio · LinkedIn · GitHub

Research Profile

Master of Computer Science student at the University of Sydney with research and engineering experience spanning multimodal learning, temporal signals, computer vision, and real-time AI systems. Experience includes robust anomaly detection in physiological time series, vision-based industrial perception, multimodal audio-visual modelling for in-cabin context, and research-facing work on language-conditioned humanoid motion. Interested in understanding how models connect visual evidence, language, time, and structured spatial context, with an emphasis on careful experimentation and reproducible evaluation.

Interests: multimodal representation learning · grounded multimodal reasoning · temporal and spatiotemporal reasoning · vision-language-action grounding · robust learning under noisy or incomplete signals · benchmark design and empirical evaluation

Education

The University of Sydney Aug 2026 - Present
Master of Computer Science (Advanced Entry) · Data Science & AI; Theory & Algorithms *Sydney, Australia*

Birla Institute of Technology and Science (BITS) Pilani Nov 2020 - Jul 2024
B.E. (Hons.) Computer Science, Minor in Data Science · CGPA 8.0/10 *Hyderabad, India*

Coursework: Machine Learning, Data Mining, Foundations of Data Science, Probability and Statistics, Signal Processing, Data Structures and Algorithms, Discrete Structures.

Research Experience

Pragya AI Lab, BITS Pilani - Goa Campus Apr 2026 - Present
Research Contributor · Supervisor: Amitava Das *Remote*

- Developed Indic-language embedding representations and alignment within a multilingual Indic embedding space, enabling direct conditioning of pretrained humanoid motion generation models on Hindi, Bengali, Tamil and Telugu without relying on Indic-to-English translation at inference.
- Worked on mapping frozen multilingual text representations into the conditioning-context geometry expected by a pretrained humanoid motion diffusion model, addressing representational mismatches that weaken language-conditioned motion generation.
- Investigated limitations of English-centric text interfaces, particularly the collapse of semantically distinct Indic instructions into identical token sequences, and their implications for instruction grounding and motion generation.
- Contributed to the PragyaVLA research codebase and developed Hugging Face Spaces for interactive access and reproducible demonstrations of the language-conditioned humanoid motion system.

BITS Pilani, Hyderabad Campus Sep 2022 - Dec 2023
Undergraduate Research Assistant · Supervisor: Prof. Chittaranjan Hota *Hyderabad, India*

- Formulated and evaluated the **Alternative Robust Mahalanobis Distance (ARMD)** for anomaly detection in physiological signals, improving detection accuracy by **25%** over the classical Mahalanobis distance.
- Designed an **LSTM autoencoder** for 144-timestep ECG windows to detect and characterise four categories of temporal anomalies, modelling structure in anomalous sequences rather than treating observations independently.
- Code: github.com/mrdetective007/UnderGrad_Research

Industry R&D Experience

APTIV Jan 2024 - July 2026
Advanced Software Engineer *Bengaluru, India*

- Investigated multimodal indicators of driver state by combining in-cabin visual cues with speech/audio, focusing on temporal and cross-modal patterns in real-world driving contexts.
- Built and evaluated a real-time speech-emotion pipeline using mel spectrograms, MFCCs, CNN/LSTM models, and Audio Spectrogram Transformers, achieving **85% accuracy on IEMOCAP**.

- Developed a VAE-based adaptive soundscape system that improved audio naturalness by approximately **20%** over static loops while maintaining **<100 ms** latency across embedded and cloud deployments.
- Engineered Android/AAOS audio infrastructure, including AIDL HAL extensions, mmap audio paths, multizone routing, and dynamic audio focus, integrating learned contextual signals with real-time system behaviour.

Larsen & Toubro Technology Services (LTTS) May 2023 - Jul 2023
Data Science Intern *Mumbai, India*

- Built a vision-based P&ID symbol-detection pipeline using MMOCR, EAST, and OpenCV, improving segmentation accuracy by **15–20%** over template-matching baselines.
- Developed semantic feature-extraction workflows linking detected symbols with their descriptions, reducing downstream human classification error by approximately **30%**.

Selected Projects

OptiRoute: TSP Algorithm Analyser Apr 2023 - Aug 2023
Python · optimisation · metaheuristics

- Implemented and systematically benchmarked seven optimisation strategies, including Genetic Algorithm, Simulated Annealing, and PSO, analysing differences in search behaviour and convergence rather than relying on a single optimisation outcome. Code

NanoGen: Mini C Compiler Jan 2023 - May 2023
C/C++ · Lex · Yacc · compiler design

- Implemented lexical, syntactic, and semantic analysis, symbol-table management, and intermediate-code generation for core C constructs, developing a complete multi-stage processing pipeline. Code

Convex Hull DCEL Jan 2023 - May 2023
Algorithms · computational geometry

- Combined convex-hull computation with a half-edge (DCEL) representation to construct and manipulate structured planar subdivisions, gaining hands-on experience with geometric representations and spatial relationships. Code

Technical Skills

Machine learning	LSTM autoencoders, CNN/LSTM models, Audio Spectrogram Transformers, VAEs, robust (Mahalanobis-based) anomaly detection
Multimodal data	Vision, speech/audio, ECG time series, temporal and cross-modal signals
Languages/frameworks	Python, C/C++, PyTorch, TensorFlow, Keras, OpenCV, MMOCR, Hugging Face Spaces
Systems	Android/AAOS audio, AIDL HAL, ALSA, embedded Linux, real-time systems

Teaching & Academic Service

The University of Sydney Aug 2026 - Dec 2026
Student Representative, CSYS5030 *Sydney, Australia*

- Represent cohort feedback to teaching staff through the formal student-representative process.

BITS Pilani, Hyderabad Campus Aug 2022 - Dec 2022
Teaching Assistant, Data Structures and Algorithms *Hyderabad, India*

- Prepared weekly lab sheets and tutorials, supported students in algorithmic problem-solving, and contributed to the design of the course's main programming assignment.

Training

CVIT Summer School	IIIT Hyderabad. Artificial Intelligence, Computer Vision and Machine Learning, Aug-Sep 2023.
Bootlin	Embedded Linux audio: ALSA, device-tree configuration, codec drivers, and embedded audio pipelines.

Additional

Languages	English, Hindi, Gujarati
Interests	Competitive table tennis (university and state level)